

MotionApps 13.1.1 for IAM-20680/20680HP/20680HT Porting Guide

SW-000000

Version 1.0

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1 GOAL OF THIS DOCUMENT

This document is intended to explain how to port the release package to the target development.

2 RELEASE PACKAGE

The release package includes following modules

- Linux kernel IIO driver for **IAM-20680/IAM-20680HP/IAM-20680HT**
 - Supports Kernel 3.18/4.4/4.9/4.9/4.14/5.4/5.10/5.15
- Android Sensors HAL 1.0
 - Supports `SENSORS_DEVICE_API_VERSION_1_4`
- Android Sensors Multi-HAL 2.1 wrapper
 - Load Sensors HAL 1.0 for implementing Sensors Multi-HAL 2.1
- Optional: AOSP 13 patch
 - Add support of Heading Sensor in AIDL wrapper
- Test applications
 - Android
 - `inv_self_test`
 - Hardware self-test and Factory Calibration
 - `test-sensorcontrol`
 - A command line application to control and stream sensor data under Android Sensor Framework
 - `invn-sensorcontrol`
 - NDK based command line application, almost the same as `test-sensorcontrol` but supports Direct Report Mode additionally

3 LINUX DRIVER

3.1 INTEGRATION

Please follow steps below.

1. Copy driver files under target Linux source tree

[Release Package]/kernel/invensense/linux-icm20xxx/files/drivers/iio/imu/inv_mpu

[Target Tree]/drivers/iio/imu/inv_mpu

Example (Kernel 4.14)

```
$ ls -l drivers/iio/imu/
total 160
-rw-r--r-- 1 eiwatsuki staff 2579 Sep 20 2019 adis16400_buffer.c
-rw-r--r-- 1 eiwatsuki staff 29181 Sep 20 2019 adis16400_core.c
-rw-r--r-- 1 eiwatsuki staff 7844 Sep 20 2019 adis16400.h
-rw-r--r-- 1 eiwatsuki staff 26478 Sep 20 2019 adis16480.c
-rw-r--r-- 1 eiwatsuki staff 4339 Sep 20 2019 adis_buffer.c
-rw-r--r-- 1 eiwatsuki staff 10084 Sep 20 2019 adis.c
-rw-r--r-- 1 eiwatsuki staff 2113 Sep 20 2019 adis_trigger.c
drwxr-xr-x 2 eiwatsuki staff 4096 Sep 20 2019 bmi160
drwxr-xr-x 3 eiwatsuki staff 4096 Sep 6 09:35 inv_mpu
drwxr-xr-x 2 eiwatsuki staff 4096 Sep 20 2019 inv_mpu6050
-rw-r--r-- 1 eiwatsuki staff 1576 Sep 20 2019 Kconfig
-rw-r--r-- 1 eiwatsuki staff 37213 Sep 20 2019 kmx61.c
-rw-r--r-- 1 eiwatsuki staff 589 Sep 20 2019 Makefile
drwxr-xr-x 2 eiwatsuki staff 4096 Sep 20 2019 st_lsm6dsx
```

2. Copy a header file under target Linux source tree

[Release Package]/hardware/invensense/linux-icm20xxx/files/include/linux/iio/imu/mpu.h

[Target Tree]/include/linux/iio/imu/mpu.h

Example (Kernel 4.14)

```
$ ls -l include/linux/iio/imu/
total 12
-rw-r--r-- 1 eiwatsuki staff 7166 Sep 20 2019 adis.h
-rw-r--r-- 1 eiwatsuki staff 3501 Sep 5 20:00 mpu.h
```

3. Apply driver integration patches

Use git am to apply integration patches located under (for Linux X.YY)

[Release Package]/hardware/invensense/linux-icm20xxx/patches/X.YY/integrate/0001-iio-imu-add-support-for-InvenSense-driver.patch

Patch example (Linux 5.4)

```
From 6e2bc7924df630ad142ed4bc224eb0c34d334316 Mon Sep 17 00:00:00 2001
From: Jean-Baptiste Maneyrol <jmaneyrol@invensense.com>
Date: Mon, 3 Sep 2018 17:35:53 +0200
Subject: [PATCH 1/3] iio: imu: add support of invensense driver

Signed-off-by: Jean-Baptiste Maneyrol <jmaneyrol@invensense.com>
---
 drivers/iio/imu/Kconfig | 1 +
 drivers/iio/imu/Makefile | 1 +
 2 files changed, 2 insertions(+)

diff --git a/drivers/iio/imu/Kconfig b/drivers/iio/imu/Kconfig
index f3c7282..eddaad9 100644
--- a/drivers/iio/imu/Kconfig
+++ b/drivers/iio/imu/Kconfig
@@ -51,6 +51,7 @@ config KMX61
     To compile this driver as module, choose M here: the module will
     be called kmx61.

+source "drivers/iio/imu/inv_mpu/Kconfig"
+source "drivers/iio/imu/inv_mpu6050/Kconfig"
+source "drivers/iio/imu/st_lsm6dsx/Kconfig"

diff --git a/drivers/iio/imu/Makefile b/drivers/iio/imu/Makefile
index 4a69588..dbfed7a 100644
--- a/drivers/iio/imu/Makefile
+++ b/drivers/iio/imu/Makefile
@@ -14,6 +14,7 @@ adis_lib-$(CONFIG_IIO_ADIS_LIB_BUFFER) +=
 adis_buffer.o
 obj-$(CONFIG_IIO_ADIS_LIB) += adis_lib.o

 obj-y += bml160/
+obj-y += inv_mpu/
+obj-y += inv_mpu6050/

 obj-$(CONFIG_KMX61) += kmx61.o
--
2.7.4
```

4. Apply kernel module fix for iio triggered buffer helper

For Linux <= 5.4, you need to apply a patch to have iio triggered buffer help compiling as a module.
[\[Release Package\]/hardware/invensense/linux-icm20xxx/patches/X.YY/0001-iio-add-Kconfig-description-for-triggered-buffer.patch](#)

Patch example (Linux 5.4)

```
From e44096f0a2c54422ce878f7b84cbbff51d7eaadd Mon Sep 17 00:00:00 2001
From: Jean-Baptiste Maneyrol <jmaneyrol@invensense.com>
Date: Fri, 4 Oct 2019 15:21:11 +0200
Subject: [PATCH] iio: add Kconfig description for triggered buffer

Required to compile the helpers in the kernel for use by an
external module.

Signed-off-by: Jean-Baptiste Maneyrol <jmaneyrol@invensense.com>
---
 drivers/iio/buffer/Kconfig | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)

diff --git a/drivers/iio/buffer/Kconfig b/drivers/iio/buffer/Kconfig
index 4ffd3db78..4dc82de66 100644
--- a/drivers/iio/buffer/Kconfig
+++ b/drivers/iio/buffer/Kconfig
@@ -37,7 +37,7 @@ config II0_KFIFO_BUF
     often to read from the buffer.

 config II0_TRIGGERED_BUFFER
-    tristate
+    tristate "Helper functions for triggered buffers"
     select II0_TRIGGER
     select II0_KFIFO_BUF
     help
--
2.17.1
```

5. Update kernel configuration (e.g. menuconfig)

Enable following options

"Industrial I/O subsystem"

"Industrial I/O buffering based on kfifo"

"Enable triggered sampling support"

"Invensense ICM20xxx devices (I2C)" or "Invensense ICM20xxx devices (SPI)"

Select **"IAM20680"** as Chip name

Example (Kernel 4.14) I2C

```
Device Drivers ->
  <*> Industrial I/O support --->
    *- Enable buffer support within IIO
    *- Industrial I/O buffering based on kfifo
    *- Enable triggered sampling support
  Inertial measurement units --->
    Chip name (IAM20680) --->
    <*> Invensense ICM20xxx devices (I2C)
```

Example (Kernel 4.14) SPI

```
Device Drivers ->
  <*> Industrial I/O support --->
    *- Enable buffer support within IIO
    *- Industrial I/O buffering based on kfifo
    *- Enable triggered sampling support
  Inertial measurement units --->
    Chip name (IAM20680) --->
    <*> Invensense ICM20xxx devices (SPI)
```

6. Update DTS file

Example is available in the release package

[Release Package]/hardware/invensense/dts/iam20680.dtsi

[Release Package]/hardware/invensense/dts/device/rb5-*.dtsi

The necessary configurations are:

1. Point a right driver

compatible = " invensense,iam20680";

2. Select voltage regulators

If required the driver can drive the 2 voltage regulators it is using (VDD for power supply and VDDIO for IOs). Fill the corresponding fields:

inven,vdd_ana-supply = <&invn_vdd>;

inven,vcc_i2c-supply = <&invn_vddio>;

3. Mount matrix

- To be used to convert from chip frame to body frame

This information will be exposed on sysfs and utilized by Sensor HAL, namely the mount matrix conversion is done in user space rather than driver.

Entries for Gyro/Accel

1. axis_map_x, axis_map_y, axis_map_z

Change axes between input and output (chip and frame).

<0> : X input

<1> : Y input

<2> : Z input

2. negate_x, negate_y, negate_z

Change the sign of the data.

<0> : no change

<1> : negate the sign

Example

axis_map_x = <1>;

axis_map_y = <0>;

axis_map_z = <2>;

negate_x = <1>;

negate_y = <0>;

negate_z = <0>;

The out will be:

X output = (-1) * Y input

Y output = X input

Z output = Z input

4. Interrupt configuration

This configuration is for the platform, not **IAM-20680** driver but it should be like below as the driver is designed to work with this configuration.

- Edge trigger
- Rising edge

Example

```
interrupts = <0 IRQ_TYPE_EDGE_RISING>;
```

5. Others

There are some configurations which are not used by **IAM-20680** but TDK InvenSense specific. Please set "none" for those entries like below.

```
inven,secondary_type = "none"
inven,aux_type = "none"
inven,read_only_slave_type = "none"
```

Example for IAM-20680 on I2C:

```
&invn_i2c {
    iam20680@68 {
        compatible = "invensense,iam20680";
        reg = <0x68>;
        interrupt-parent = <&invn_gpio>;
        interrupts = <0 IRQ_TYPE_EDGE_RISING>;
        inven,vdd_ana-supply = <&invn_vdd>;
        inven,vcc_i2c-supply = <&invn_vddio>;
        axis_map_x = <1>;
        axis_map_y = <0>;
        axis_map_z = <2>;
        negate_x = <1>;
        negate_y = <0>;
        negate_z = <0>;
        inven,secondary_type = "none";
        inven,aux_type = "none";
        inven,read_only_slave_type = "none";
    };
};
```

Example for IAM-20680 on SPI:

```
&invn_spi {
    iam20680@0 {
        compatible = "invensense,iam20680";
        reg = <0>;
        interrupt-parent = <&invn_gpio>;
        interrupts = <0 IRQ_TYPE_EDGE_RISING>;
        inven,vdd_ana-supply = <&invn_vdd>;
        inven,vcc_i2c-supply = <&invn_vddio>;
        axis_map_x = <1>;
        axis_map_y = <0>;
        axis_map_z = <2>;
        negate_x = <1>;
        negate_y = <0>;
        negate_z = <0>;
        inven,secondary_type = "none";
        inven,aux_type = "none";
        inven,read_only_slave_type = "none";
    };
};
```

4 ANDROID INTEGRATION

4.1 AOSP PATCHES

This step is optional and only needed for AOSP 13 Heading Sensor support.

Until the following issue is fixed and integrated: <https://issuetracker.google.com/issues/263462499>

To apply the patch, do the following:

- `cp [Release Package]/patches/hardware/invensense/0001-Add-missing-HEADING-sensor-support.patch [Target Tree]/hardware/interfaces/`
- `cd [Target Tree]/hardware/interfaces/`
- `git apply 0001-Add-missing-HEADING-sensor-support.patch`

If you don't need Heading sensor, you can delete it by commenting the following lines:

hardware/invensense/sensors/SensorsList.h

```
/* {"Invensense Heading", "Invensense", 1,
   SENSORS_HEADING_HANDLE,
   SENSOR_TYPE_HEADING, 359.99f, 0.01f, 0.5f, 5000, 0, 0,
   SENSOR_STRING_TYPE_HEADING, "", 250000,
   SENSOR_FLAG_CONTINUOUS_MODE, {}}, */
```

4.2 SENSORS HAL

There are several steps to follow.

1. Copy files to under Android source tree
[Release Package]/aosp/hardware/invensense
[Target Tree]/hardware
2. Update device files (init & ueventd)
Please refer to *hardware/invensense/sensors/integration/README* for more information.
3. Update sepolicy
This would be required to allow Sensors HAL to access sysfs entries.
Please refer to *hardware/invensense/sensors/integration/README* for more information.
4. Integrate Sensors HAL 1.0 in device file and set Sensors HAL library name

```
# InvenSense Sensors HAL
PRODUCT_PACKAGES += sensors.iam20680
PRODUCT_PROPERTY_OVERRIDES += ro.hardware.sensors=iam20680
```

Final integration depends on the Android version and the Sensors HAL version supported.

1. Sensors HAL 1.0: compatible with all Android <= 13

```
# Sensors HAL 1.0 integration
PRODUCT_PACKAGES += \
    android.hardware.sensors@1.0-impl \
    android.hardware.sensors@1.0-service
```

2. Sensors Multi-HAL 2.1: compatible with Android 11, 12, 13

```
# Sensors Multi-HAL 2.1 integration
PRODUCT_PACKAGES += \
    android.hardware.sensors@2.1-multihal-invensense \
    android.hardware.sensors@2.1-service.multihal
```

Sensors Multi-HAL usage requires to copy a **hals.conf** file under **/vendor/etc/sensors/** listing all the Sensors HALs to load.

InvenSense HAL name to use: **android.hardware.sensors@2.1-multihal-invensense.so**

3. Sensors Multi-HAL AIDL: compatible with Android 13

```
# Sensors Multi-HAL AIDL integration
PRODUCT_PACKAGES += \
    android.hardware.sensors@2.1-multihal-invensense \
    android.hardware.sensors-service.multihal
```

Sensors Multi-HAL usage requires to copy a **hals.conf** file under **/vendor/etc/sensors/** listing all the Sensors HALs to load.

InvenSense HAL name to use: **android.hardware.sensors@2.1-multihal-invensense.so**

5 CALIBRATION & TEST APPLICATIONS FOR ANDROID

Add following lines in the device file to build the calibration and test applications within AOSP.

```
# InvenSense tools
PRODUCT_PACKAGES += \
    inv_self_test \
    test-sensorcontrol \
    invn-sensorcontrol
```

5.1 INV_SELF_TEST

[Release Package]/aosp/hardware/invensense/sensors/tools/self_test

The usage will be shown when running this application with “-h” option.

The default behavior is to run chip self test, do factory calibration and write new calibration file with factory calibration.

5.2 TEST-SENSORCONTROL

[Release Package]/aosp/hardware/invensense/tools/sensorcontrol

The usage will be shown when running this application with “-h” option.

This tool can only be built inside AOSP integrated inside the image.

5.3 INVN-SENSORCONTROL

[Release Package]/aosp/hardware/invensense/tools/invn-sensorcontrol

The usage will be shown when running this application with “-h” option.

This tool can be built independently outside of AOSP using NDK ndk-build tool.

6 DOCUMENT INFORMATION

6.1 REVISION HISTORY

Date	Revision	Description
05/26/2023	1.0	Initial release